

Task 5

Exactly. Keep that as the **advanced task for later**.

● Advanced Spark Structured Streaming Task

You'll build a realistic streaming pipeline covering:

1. **Joins**
 - Stream-to-static join
 - Stream-to-stream join
2. **Stateful operations**
 - Maintain state across batches
 - Stateful aggregations
3. **Deduplication**
 - Remove duplicate events
 - Deduplicate using event IDs
4. **Late events**
 - Event-time processing
 - Watermarks
 - Handling events that arrive late
5. **foreachBatch**
 - Process each micro-batch
 - Write/update multiple sinks
 - Use batch-level logic
6. **Output**
 - JSON input
 - Transformations
 - Parquet output
 - Checkpointing
 - HDFS

Difficulty progression

- Beginner → Read JSON → Transform → Write Parquet/JSON
- Intermediate → Aggregations → Windows → Watermarks → Multiple outputs
- Advanced → **Joins + Stateful operations + Deduplication + Late events + foreachBatch**

When we reach this task, I'll make it a **complete realistic project**, including the JSON data files and the full PySpark code step by step.